

Response to Reviewers

We thank the reviewers for their constructive and encouraging comments. The manuscript has been revised substantially, with the aim of making the contribution clearer, more complete and easier to read. In particular, we clarified the main message of the paper, added a flow-oriented description of the proposed data-to-information process, strengthened the distinction between ML/XAI processing and LLM-assisted reporting, expanded the discussion of user-tailored air-quality information, and corrected typographical, referencing and formatting issues.

Reviewer 1

1. Flow-oriented presentation of data fusion, information exploration and spatial assessment.

We agree that the original version would benefit from a clearer process view. The revised manuscript adds a flow-oriented overview of the information maturation chain and reorganises the technical discussion around the steps from heterogeneous data sources to integration policy, semantic harmonisation, QA/QC, spatial ML assessment, XAI-supported exploration and LLM-assisted information products. This makes the role of each processing point more explicit.

2. Central message of the manuscript.

We added an abstract and we revised the Introduction to state the central message more directly: ML supports the fusion of heterogeneous spatial evidence and the production of forecasts, uncertainty descriptors and explanation objects, while the LLM layer acts as a controlled information interface that transforms validated evidence into audience-specific reports. The paper is now explicitly framed as an implementation-oriented environmental-informatics specification rather than as a single experimental demonstrator.

3. Data integration, data warehouses and distributed/federated access.

We added material on integration policies for heterogeneous environmental streams. The revised text distinguishes between centralised storage, data-warehouse-style integration and distributed or federated access. It also clarifies that different data types require different handling policies: high-frequency IoT streams, satellite/model products, CityGML/OGC urban layers and vulnerability or population layers differ in latency, spatial support, ownership, uncertainty and metadata needs.

4. Harmonisation and ML role.

We clarified that harmonisation is primarily metadata- and semantics-driven, while ML is applied later for calibration, imputation, feature selection, fusion, forecasting and explainability. The revised manuscript therefore separates semantic integration from model training and avoids implying that ML alone performs the harmonisation task.

5. Minor issues.

We corrected the reported missing cross-reference and typographical problems, checked the final LaTeX build, and verified that the final PDF compiles without undefined references or citations.

Reviewer 2

1. Mathematical formalisation.

We clarified the purpose of the notation. The mathematical expressions are not intended for theorem proving, but as a compact way to define data objects, spatial supports, transformations, evidence packages and information products in a reproducible workflow. We also streamlined the text around the equations where possible.

2. Specification rather than complete demonstrator.

We agree that the manuscript should not overstate its implementation status. The Introduction, Evaluation and Discussion now make clear that the contribution is an implementation-oriented framework specification. The revised Evaluation section defines the validation and reproducibility dimensions that an operational deployment should document, while full benchmarking across cities, pollutants, sensor networks and LLM versions is explicitly left for future deployment work.

3. Position of glossary and small corrections.

The list of symbols and abbreviations was moved after the introductory problem statement, so that readers first encounter the motivation and contribution. We also added missing abbreviations and corrected the wording issues noted by the reviewer, including wording such as structured, framework treats, components, degree and its enrichment.

4. Transferability.

We added a discussion of transferability to related environmental and risk-information domains, including disaster management, smart agriculture and smart forestry. The revised discussion emphasises that the same architecture is relevant wherever heterogeneous data integration, spatial assessment, uncertainty handling and audience-specific communication are required.

Reviewer 3

1. SHAP and LIME.

We expanded the XAI discussion and the abbreviations table. SHAP is now described as a method for global and local feature-attribution analysis, while LIME is described as a local surrogate-explanation method for individual predictions. The revised ML/XAI section also explains how XAI can support feature reduction, model interpretation and the production of explanation objects for later reporting.

2. Benefits of the LLM layer and inadequacies of current air-quality information.

We added a dedicated discussion explaining that many air-quality services provide maps, time series, exceedance indicators or raw forecasts, but often remain too technical for citizens, too fragmented for decision makers and insufficiently tailored for sensitive groups. The revised Section 5 connects this gap to earlier work on personalised environmental information, ontology-based recommendations and multimodal air-quality communication, and then explains how the LLM layer converts the same validated evidence package into different communication products.

3. Choice of LLM and possible use of smaller models.

We clarified that the framework is model-agnostic. It may use a hosted LLM, a locally deployed open model or a domain-adapted small language model, provided that requirements for faithfulness, latency, privacy, cost, language coverage and controllability are met. The illustrative Gemini 2.0 Flash report is presented only as an example of the reporting layer, not as a prescribed model choice.

Additional final edits

In the final 18-page version we also corrected the spelling of considered, harmonised terminology, improved the wording on spatial supports. These final edits improve clarity and compactness while preserving the revised manuscript's technical scope.